

Buddy Challenge Part 5

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Loading the data and labeling

```
#loading in the data
data <- read.csv('file_with_data_without_outliers_accuracies.csv')

#labeling factors
data$flanker_condition <- factor(data$flanker_condition,
labels=c("unrelated", "relatedn-3", "relatedn-2", "relatedn-1", "relatedn+1",
↵ "relatedn+2", "relatedn+3"))

data$flanker_condition <- relevel(data$flanker_condition, ref="unrelated")
```

Setting our model formula

```
#Model formula
model_formula <- bf(correct_keyboard_response ~ flanker_condition +
(1+flanker_condition|Participant_unique_ID) +
(1+flanker_condition|target))
```

Setting priors

```
# get priors
get_prior(model_formula, data)

# Specifying priors
my_priors <- c(prior(normal(0, 1), class = 'Intercept'),
prior(normal(0, 1), class = 'b'))
```

Running model the model

```
# Run model
model_weak <- brm(correct_keyboard_response ~ flanker_condition
  + (1+flanker_condition|Participant_unique_ID)
  + (1+flanker_condition|target),
  family = bernoulli(link = "logit"),
  prior = my_priors,
  data = data,
  chains = 4,
  cores = 4,
  iter = 2000,
  warmup = 1000,
  thin = 1,
  seed = 123)

summary(model_weak)
```

Defining SESOI

We did a literature search to define the SESOI for accuracy measurements. Unfortunately, we were not able to find any suitable previous research to justify our choice. The most relevant literature we were able to find was research on the psychometric function, where the rate for stimulus-independent errors in trained observers was found to be between 0% and 5% [Wichmann 2001 psychometric]. Thus, we decided the smallest effect size of interest to be a difference of 5% in probability space, formalizing our ROPE as -5% to 5% from point 0 (converted to log odds). As we did not find literature to confidently define our SESOI, we also want to investigate the smallest measurable change, meaning change in one trial. This corresponds to 0.005%.

```
# define ROPE - wide rope of 5%
rope <- c(qlogis(0.45), qlogis(0.55))

# defining a smaller rope of 0.005% (0.0045%), equivalent to one trial change
rope_small <- c(qlogis(0.49995), qlogis(0.50005))
```

Bayes factor 5%

```
BF_5_M3 <-
  bayesfactor_rope(posterior = model_weak,
    null = rope,
    parameter = "b_flanker_conditionrelatednM3")

BF_5_M3
# 0.028

BF_5_M2 <-
  bayesfactor_rope(posterior = model_weak,
    null = rope,
    parameter = "b_flanker_conditionrelatednM2")
```

```

BF_5_M2
# 0.152

BF_5_M1 <-
  bayesfactor_rope(posterior = model_weak,
                    null = rope,
                    parameter = "b_flanker_conditionrelatednM1")

BF_5_M1
# 0.098

BF_5_P3 <-
  bayesfactor_rope(posterior = model_weak,
                    null = rope,
                    parameter = "b_flanker_conditionrelatednP3")

BF_5_P3
# 0.014

BF_5_P2 <-
  bayesfactor_rope(posterior = model_weak,
                    null = rope,
                    parameter = "b_flanker_conditionrelatednP2")

BF_5_P2
# 0.054

BF_5_P1 <-
  bayesfactor_rope(posterior = model_weak,
                    null = rope,
                    parameter = "b_flanker_conditionrelatednP1")

BF_5_P1
# 0.066

```

Running sensitivity analysis for different priors

```

#setting prior for intercept
priors <- c(prior(normal(0, 1), class = 'Intercept'))

# defining more priors
priors_0.5 <- c(priors,
               prior(normal(0, 0.5), class = b))
priors_0.75 <- c(priors,
                prior(normal(0, 0.75), class = b))
priors_1 <- c(priors,
             prior(normal(0, 1), class = b))
priors_1.25 <- c(priors,

```

```

        prior(normal(0, 1.25), class = b))
priors_1.5 <- c(priors,
               prior(normal(0, 1.5), class = b))

# Defining a list of these prior specifications
prior_list <- list(priors_0.5,
                  priors_0.75,
                  priors_1,
                  priors_1.25,
                  priors_1.5
)

# Initialize a list to store models
model_list <- list()

# Loop over the priors
# for (i in seq_along(prior_list)) {
#   model_list[[i]] <-
#     brm(
#       formula = model_formula,
#       data = data,
#       prior = prior_list[[i]],
#       # common sampling specifications
#       seed = 123,
#       iter = 4000,
#       chains = 4,
#       cores = 4,
#       backend = "cmdstanr"
#     )
# }

# name models
names(model_list) <- paste0("model_with_priors", c(0.5, 0.75, 1, 1.25, 1.5))

# saveRDS(model_list, "model_priors.RDS")
model_list_all <- readRDS("model_priors.RDS")

```

Loop through different ROPEs

```

# specify different rope intervals
ropes <- list(c(qlogis(0.45), qlogis(0.55)),
              c(qlogis(0.47), qlogis(0.53)),
              c(qlogis(0.49), qlogis(0.51)),
              c(qlogis(0.49995), qlogis(0.50005)))

# m1

```

```

results <- map_dfr(names(model_list_all), function(model_name) {
  model <- model_list_all[[model_name]]

  map_dfr(ropes, function(interval) {
    lower <- interval[1]
    upper <- interval[2]

    bayesfactor_parameters(
      model,
      null = c(lower, upper),
      parameter = "b_flanker_conditionrelatednM1"
    ) %>%
      as_tibble() %>%
      mutate(
        model = model_name,
        interval_range = paste0("[", lower, ", ", upper, "]")
      ) %>%
      select(model, everything())
  })
})

# wrangle
results_m1 <- results |>
  filter(Parameter == "b_flanker_conditionrelatednM1") |>
# separate(model, sep = "_", into = c(NA, NA, NA, NA, "prior_sd")) |>
  mutate(BF = exp(log_BF)) |>
  mutate_if(is.numeric, round, digits = 3)|>
  select(model, BF, interval_range)

#m3
results_m3 <- map_dfr(names(model_list_all), function(model_name) {
  model <- model_list_all[[model_name]]

  map_dfr(ropes, function(interval) {
    lower <- interval[1]
    upper <- interval[2]

    bayesfactor_parameters(
      model,
      null = c(lower, upper),
      parameter = "b_flanker_conditionrelatednM3"
    ) %>%
      as_tibble() %>%
      mutate(
        model = model_name,
        interval_range = paste0("[", lower, ", ", upper, "]")
      ) %>%
      select(model, everything())
  })
})

```

```

})
})

# wrangle
results_m3 <- results_m3 |>
  filter(Parameter == "b_flanker_conditionrelatednM3") |>
  # separate(model, sep = "_", into = c(NA, NA, NA, NA, "prior_sd")) |>
  mutate(BF = exp(log_BF)) |>
  # mutate_if(is.numeric, round, digits = 3)|>
  select(model, BF, interval_range)

#m2
results_m2_1 <- map_dfr(names(model_list_all), function(model_name) {
  model <- model_list_all[[model_name]]

  map_dfr(ropes, function(interval) {
    lower <- interval[1]
    upper <- interval[2]

    bayesfactor_parameters(
      model,
      null = c(lower, upper),
      parameter = "b_flanker_conditionrelatednM2"
    ) %>%
    as_tibble() %>%
    mutate(
      model = model_name,
      interval_range = paste0("[", lower, ", ", upper, "]")
    ) %>%
    select(model, everything())
  })
})

# wrangle
results_m2 <- results_m2_1 |>
  filter(Parameter == "b_flanker_conditionrelatednM2") |>
  # separate(model, sep = "_", into = c(NA, NA, NA, NA, "prior_sd")) |>
  mutate(BF = exp(log_BF)) |>
  mutate_if(is.numeric, round, digits = 3)|>
  select(model, BF, interval_range)

#p1
results_p1_1 <- map_dfr(names(model_list_all), function(model_name) {
  model <- model_list_all[[model_name]]

  map_dfr(ropes, function(interval) {

```

```

lower <- interval[1]
upper <- interval[2]

bayesfactor_parameters(
  model,
  null = c(lower, upper),
  parameter = "b_flanker_conditionrelatednP1"
) %>%
  as_tibble() %>%
  mutate(
    model = model_name,
    interval_range = paste0("[", lower, ", ", upper, "]")
  ) %>%
  select(model, everything())
})

# wrangle
results_p1 <- results_p1_1 |>
  filter(Parameter == "b_flanker_conditionrelatednP1") |>
# separate(model, sep = "_", into = c(NA, NA, NA, NA, "prior_sd")) |>
mutate(BF = exp(log_BF)) |>
  mutate_if(is.numeric, round, digits = 3)|>
  select(model, BF, interval_range)

#p2
results_p2 <- map_dfr(names(model_list_all), function(model_name) {
  model <- model_list_all[[model_name]]

  map_dfr(ropes, function(interval) {
    lower <- interval[1]
    upper <- interval[2]

    bayesfactor_parameters(
      model,
      null = c(lower, upper),
      parameter = "b_flanker_conditionrelatednP2"
    ) %>%
      as_tibble() %>%
      mutate(
        model = model_name,
        interval_range = paste0("[", lower, ", ", upper, "]")
      ) %>%
      select(model, everything())
    })
})

# wrangle

```

```

results_p2 <- results_p2 |>
  filter(Parameter == "b_flanker_conditionrelatednP2") |>
# separate(model, sep = "_", into = c(NA, NA, NA, NA, "prior_sd")) |>
  mutate(BF = exp(log_BF)) |>
  mutate_if(is.numeric, round, digits = 3)|>
  select(model, BF, interval_range)

#p3
results_p3_1 <- map_dfr(names(model_list_all), function(model_name) {
  model <- model_list_all[[model_name]]

  map_dfr(ropes, function(interval) {
    lower <- interval[1]
    upper <- interval[2]

    bayesfactor_parameters(
      model,
      null = c(lower, upper),
      parameter = "b_flanker_conditionrelatednP3"
    ) %>%
      as_tibble() %>%
      mutate(
        model = model_name,
        interval_range = paste0("[", lower, ", ", upper, "]")
      ) %>%
      select(model, everything())
  })
})

# wrangle
results_p3 <- results_p3_1 |>
  filter(Parameter == "b_flanker_conditionrelatednP3") |>
# separate(model, sep = "_", into = c(NA, NA, NA, NA, "prior_sd")) |>
  mutate(BF = exp(log_BF)) |>
  mutate_if(is.numeric, round, digits = 3)|>
  select(model, BF, interval_range)

```

Figures - Bayes factor in favor of alternative

```

# plot m3

m3 <- ggplot(results_m3,
  aes(x = as.factor(model),
      y = as.factor(interval_range),
      fill = BF)) +
  geom_tile(colour = "grey") +
  geom_text(aes(label = round(BF,3))) +
  scale_fill_gradient2(transform = "log",

```

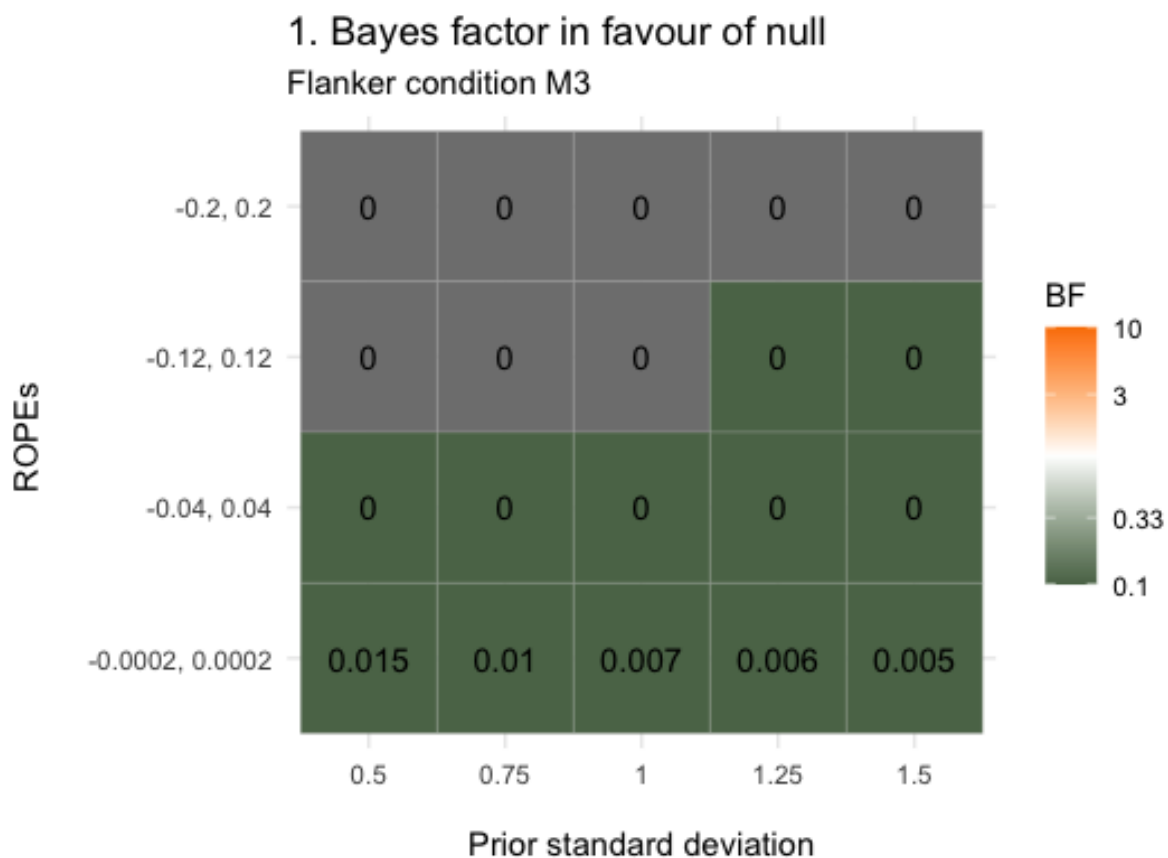


```

limits = c(0.1,10),
midpoint = 1,
low = "#5C7457",
high = "#FA8100",
breaks = c(0.00000,0.03,0.1,0.33,0.3,10),
labels = c("0","0.03","0.1","0.33","1","3","10"),
oob = scales::squish
) +
scale_y_discrete(labels = c("-0.0002, 0.0002", "-0.04, 0.04", "-0.12, 0.12",
↪ "-0.2, 0.2")) +
scale_x_discrete(labels = c("0.5", "0.75", "1", "1.25", "1.5")) +
labs(title = "1. Bayes factor in favour of null",
      subtitle = "Flanker condition M3",
      fill = "BF",
      y = "ROPEs \n ",
      x = "\nPrior standard deviation") +
theme_minimal()

```

```
print(m3)
```



```

# plot m2

m2 <- ggplot(results_m2,
  aes(x = as.factor(model),

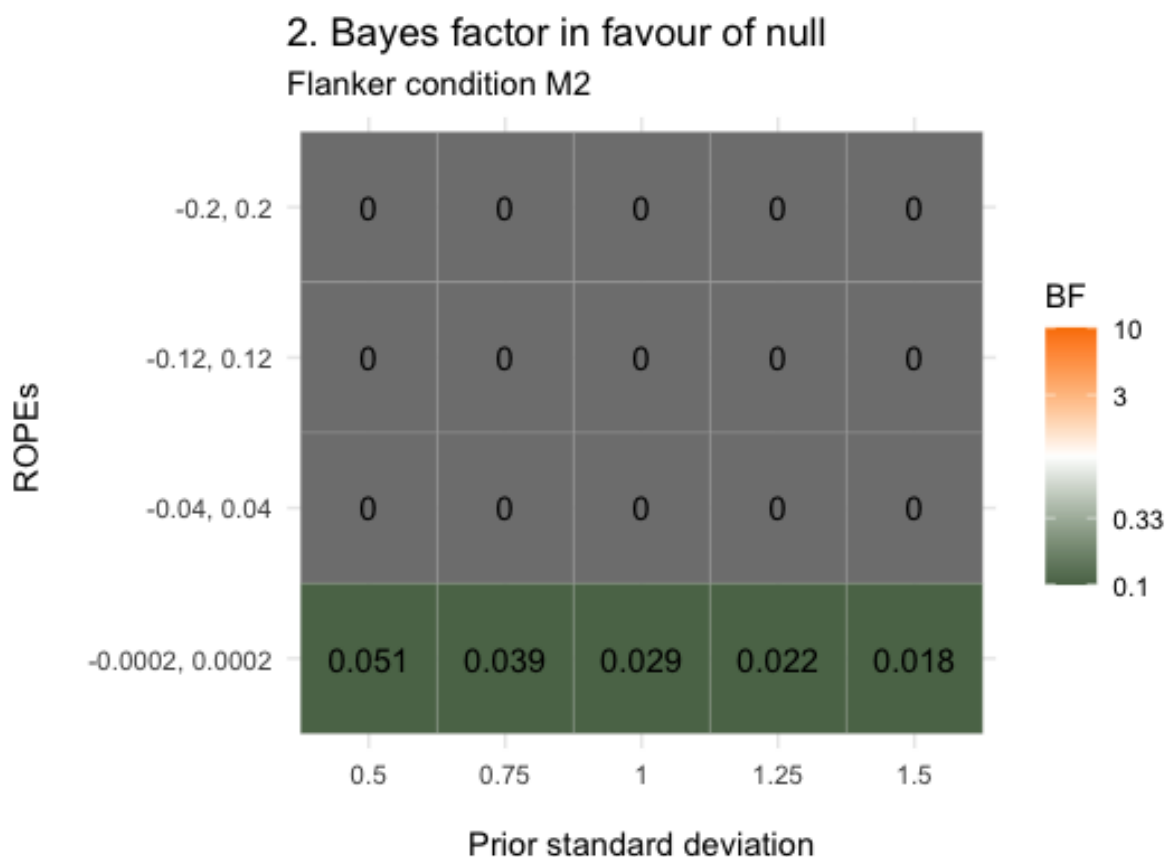
```

```

      y = as.factor(interval_range),
      fill = BF)) +
geom_tile(colour = "grey") +
geom_text(aes(label = round(BF,3))) +
scale_fill_gradient2(transform = "log",
                      limits = c(0.1,10),
                      midpoint = 1,
                      low = "#5C7457",
                      high = "#FA8100",
                      breaks = c(0.00000,0.03,0.1,0.33,0.3,10),
                      labels = c("0","0.03","0.1","0.33","1","3","10"),
                      oob = scales::squish
                      ) +
scale_y_discrete(labels = c("-0.0002, 0.0002", "-0.04, 0.04", "-0.12, 0.12",
  ↪  "-0.2, 0.2")) +
scale_x_discrete(labels = c("0.5", "0.75", "1", "1.25", "1.5")) +
labs(title = "2. Bayes factor in favour of null",
     subtitle = "Flanker condition M2",
     fill = "BF",
     y = "ROPEs \n ",
     x = "\nPrior standard deviation") +
theme_minimal()

print(m2)

```



```

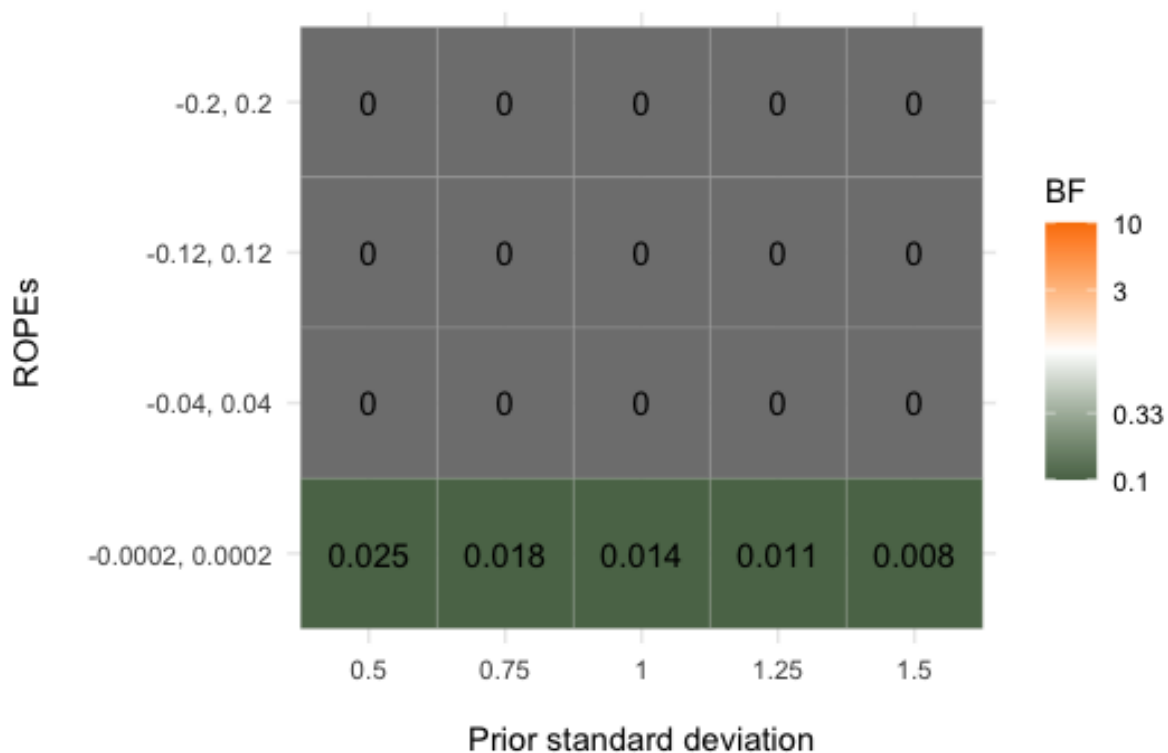
# plot m1

m1 <- ggplot(results_m1,
  aes(x = as.factor(model),
      y = as.factor(interval_range),
      fill = BF)) +
  geom_tile(colour = "grey") +
  geom_text(aes(label = round(BF,3))) +
  scale_fill_gradient2(transform = "log",
    limits = c(0.1,10),
    midpoint = 1,
    low = "#5C7457",
    high = "#FA8100",
    breaks = c(0.00000,0.03,0.1,0.33,0.3,10),
    labels = c("0","0.03","0.1","0.33","1","3","10"),
    oob = scales::squish
  ) +
  scale_y_discrete(labels = c("-0.0002, 0.0002", "-0.04, 0.04", "-0.12, 0.12",
    ↵ "-0.2, 0.2")) +
  scale_x_discrete(labels = c("0.5", "0.75", "1", "1.25", "1.5")) +
  labs(title = "3. Bayes factor in favour of null",
    subtitle = "Flanker condition M1",
    fill = "BF",
    y = "ROPEs \n ",
    x = "\nPrior standard deviation") +
  theme_minimal()
print(m1)

```

3. Bayes factor in favour of null

Flanker condition M1



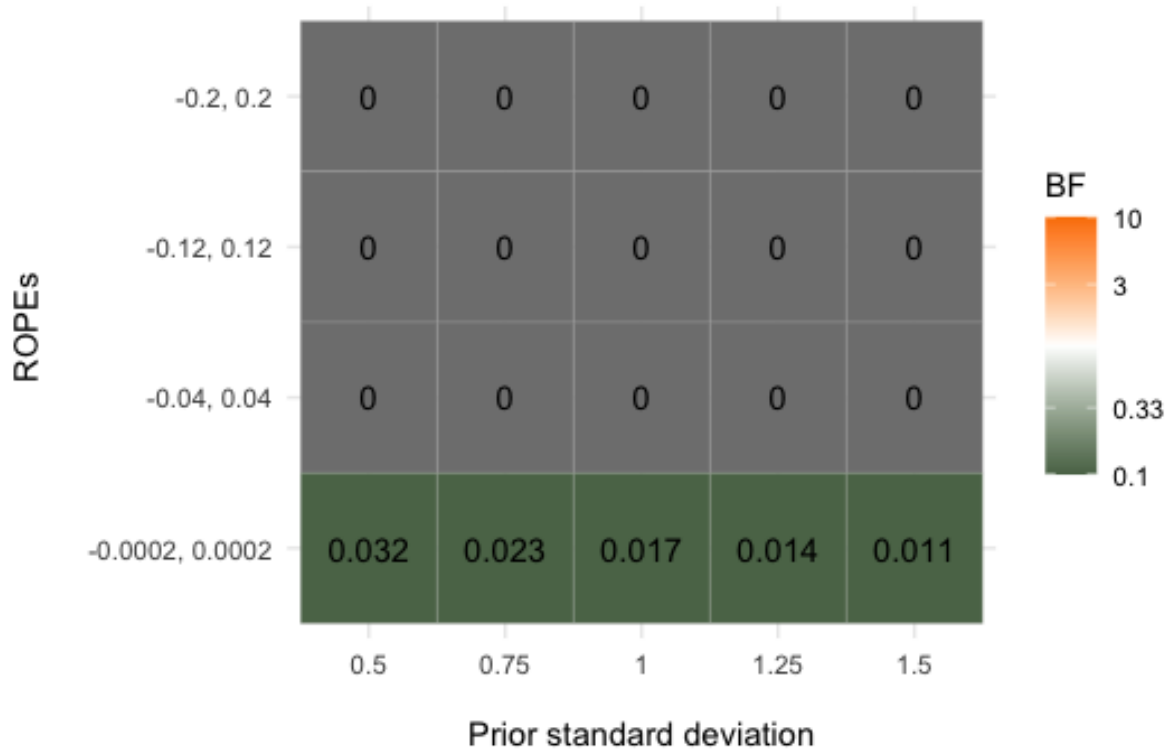
```
# plot p1

p1 <- ggplot(results_p1,
  aes(x = as.factor(model),
      y = as.factor(interval_range),
      fill = BF)) +
  geom_tile(colour = "grey") +
  geom_text(aes(label = round(BF,3))) +
  scale_fill_gradient2(transform = "log",
    limits = c(0.1,10),
    midpoint = 1,
    low = "#5C7457",
    high = "#FA8100",
    breaks = c(0.00000,0.03,0.1,0.33,0.3,10),
    labels = c("0","0.03","0.1","0.33","1","3","10"),
    oob = scales::squish
  ) +
  scale_y_discrete(labels = c("-0.0002, 0.0002", "-0.04, 0.04", "-0.12, 0.12",
    ↵ "-0.2, 0.2")) +
  scale_x_discrete(labels = c("0.5", "0.75", "1", "1.25", "1.5")) +
  labs(title = "4. Bayes factor in favour of null",
    subtitle = "Flanker condition P1",
    fill = "BF",
    y = "ROPEs \n ",
    x = "\nPrior standard deviation") +
```

```
theme_minimal()
print(p1)
```

4. Bayes factor in favour of null

Flanker condition P1



```
# plot p2

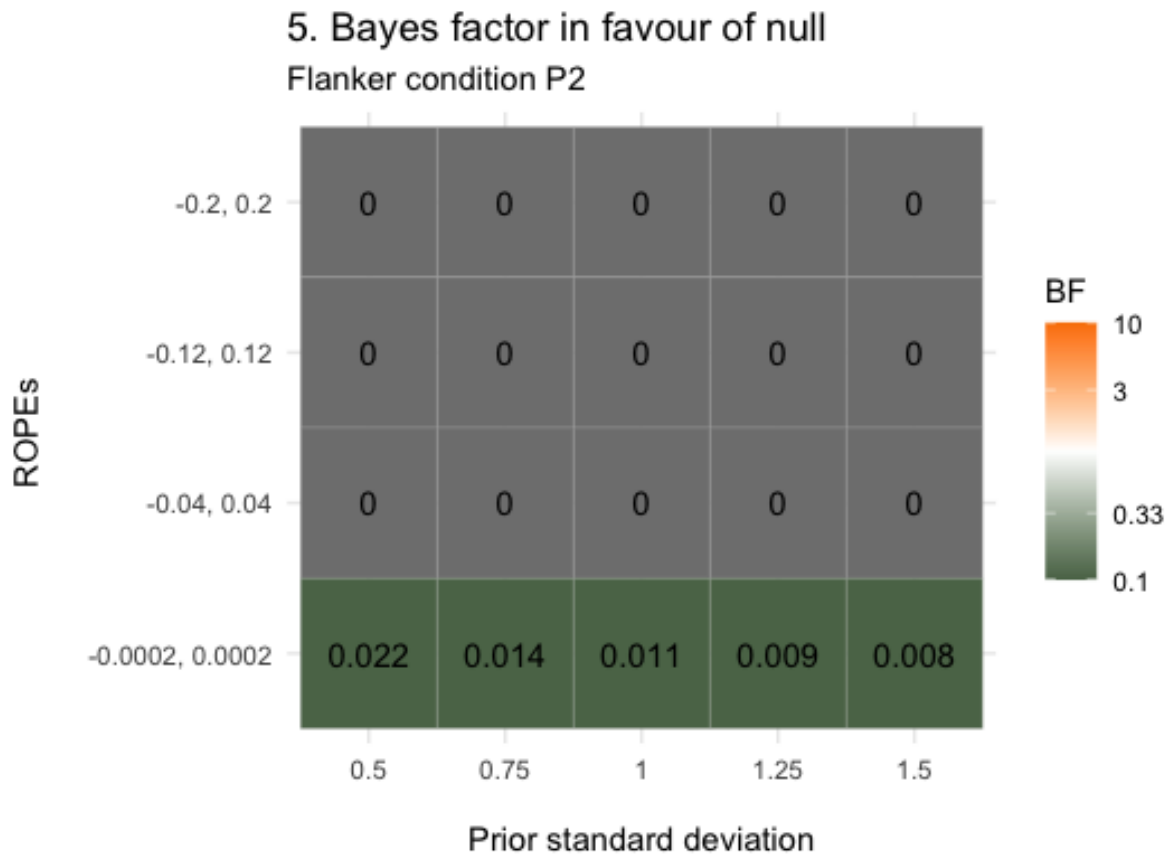
p2 <- ggplot(results_p2,
  aes(x = as.factor(model),
      y = as.factor(interval_range),
      fill = BF)) +
  geom_tile(colour = "grey") +
  geom_text(aes(label = round(BF,3))) +
  scale_fill_gradient2(transform = "log",
    limits = c(0.1,10),
    midpoint = 1,
    low = "#5C7457",
    high = "#FA8100",
    breaks = c(0.00000,0.03,0.1,0.33,0,3,10),
    labels = c("0","0.03","0.1","0.33","1","3","10"),
    oob = scales::squish
  ) +
  scale_y_discrete(labels = c("-0.0002, 0.0002", "-0.04, 0.04", "-0.12, 0.12",
    ↵ "-0.2, 0.2")) +
  scale_x_discrete(labels = c("0.5", "0.75", "1", "1.25", "1.5")) +
  labs(title = "5. Bayes factor in favour of null",
```

```

    subtitle = "Flanker condition P2",
    fill = "BF",
    y = "ROPEs \n ",
    x = "\nPrior standard deviation") +
  theme_minimal()

print(p2)

```



```

# plot p3

p3 <- ggplot(results_p3,
  aes(x = as.factor(model),
    y = as.factor(interval_range),
    fill = BF)) +
  geom_tile(colour = "grey") +
  geom_text(aes(label = round(BF,3))) +
  scale_fill_gradient2(transform = "log",
    limits = c(0.1,10),
    midpoint = 1,
    low = "#5C7457",
    high = "#FA8100",
    breaks = c(0.00000,0.03,0.1,0.33,0,3,10),
    labels = c("0","0.03","0.1","0.33","1","3","10"),
    oob = scales::squish

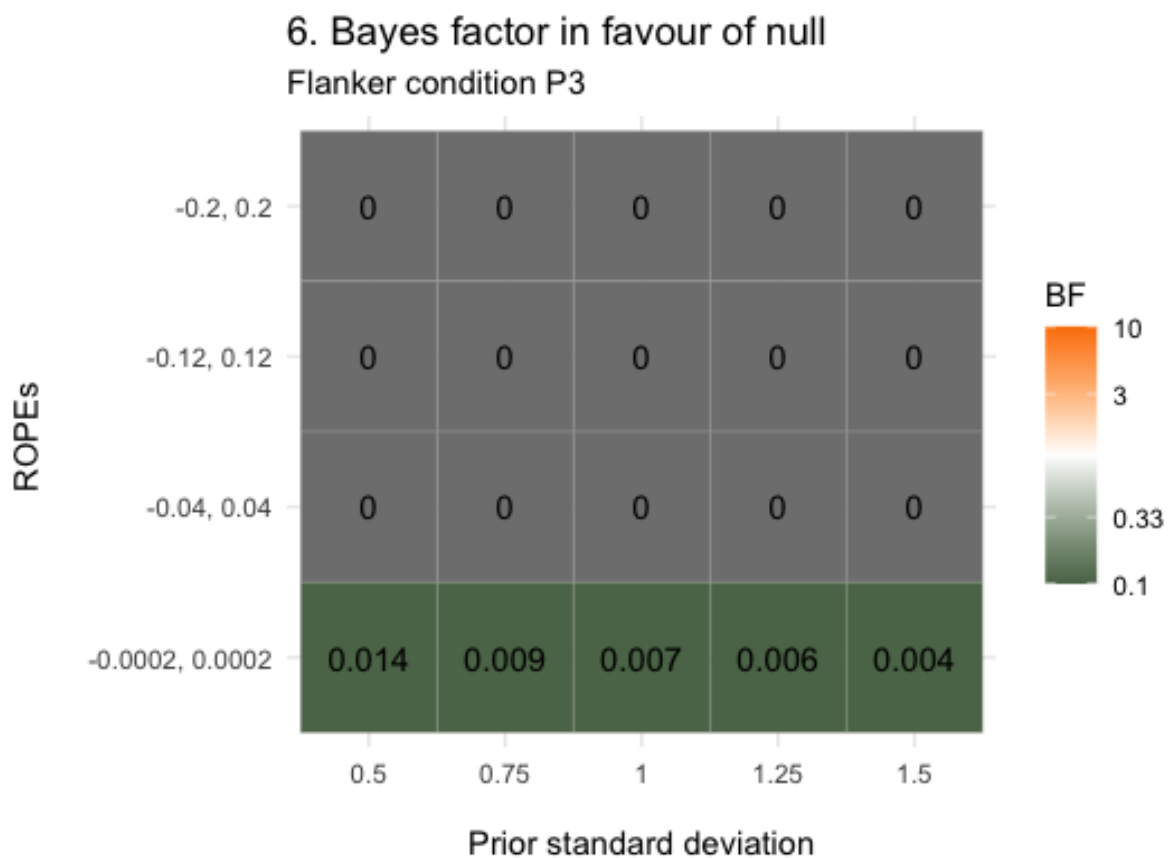
```

```

    ) +
scale_y_discrete(labels = c("-0.0002, 0.0002", "-0.04, 0.04", "-0.12, 0.12",
  ↵ "-0.2, 0.2")) +
scale_x_discrete(labels = c("0.5", "0.75", "1", "1.25", "1.5")) +
labs(title = "6. Bayes factor in favour of null",
     subtitle = "Flanker condition P3",
     fill = "BF",
     y = "ROPEs \n ",
     x = "\nPrior standard deviation") +
theme_minimal()

print(p3)

```



Write up

The data were modeled using a mixed-effect model predicting accuracy by flanker condition (the placement of the target word) and includes by-subject random slopes ($n = 98$) and by-target random slopes ($n = x$). The model was fit using brms (Bürkner, 2018) in R (R Core Team, 2025). We used default priors, estimated from the data by brms, and specified weakly informative priors for the accuracy (normal, mean = 0, sd = 1), and specified reasonably weakly informative priors for the predictor of interest: flanker conditions (normal, mean = 0, sd = [0.25, 0.5, 1, 1.25, 1.5]). We fit this model with four chains of sampling for the estimation of the joint posterior distribution, and 4000 iterations with 2000 for warm-up per chain.

Using the Bayesian framework, we seek to evaluate the evidence for a meaningful effect of flanker condition on accuracy based on the data against the background of our model and chosen priors. We define a region of practical equivalence (ROPE) that represents a reasonable change in accuracy around zero that we consider to not be meaningful (Kruschke, 2018). We decided the smallest effect size of interest to be a difference of 5% in probability space, formalizing our ROPE as -5% to 5% from point 0 (converted to log odds). We assumed ROPE intervals centered on zero from the smallest possible change (change in accuracy in one trial), 0.005% to 5%, with two middle intervals of 1% and 3%.

From this, we consider each pair of ROPE and prior. The overall conclusion from the sensitivity analysis is that there is extreme evidence for the null hypothesis. See Figures 1-6. With the most narrow ROPE, the evidence for H0 is between strong (below 0.1), very strong (below 0.03) and extreme (below 0.01). The wider the prior the lower the Bayes factors. By exploring a variety of prior choices, the conclusion does not change. We are still finding that flanker condition has no effect on accuracy.

References

```
@article{wichmann2001psychometric,
  title={The psychometric function: I. Fitting, sampling, and goodness of fit}, author={Wichmann, Felix A and Hill, N Jeremy}, journal={Perception & psychophysics},
  volume={63},
  number={8},
  pages={1293--1313},
  year={2001},
  publisher={Springer} }
```

```
@article{kruschke2018rejecting,
  title={Rejecting or accepting parameter values in Bayesian estimation},
  author={Kruschke, John K},
  journal={Advances in methods and practices in psychological science},
  volume={1},
  number={2},
  pages={270–280},
  year={2018},
  publisher={Sage Publications Sage CA: Los Angeles, CA} }
```

```
@misc{RN305,
  author = {R Core Team},
  title = {R: A Language and Environment for Statistical Computing},
  publisher = {R Foundation for Statistical Computing},
  url = {https://www.R-project.org/},
  year = {2023},
  type = {Generic} }
```

```
@article{RJ-2018-017,
  author = {Bürkner, Paul-Christian},
  title = {Advanced Bayesian Multilevel Modeling with the R Package brms},
  journal = {The R Journal},
  year = {2018},
  note = {https://doi.org/10.32614/RJ-2018-017},
  doi = {10.32614/RJ-2018-017},
  volume = {10},
  issue = {1},
```


issn = {2073-4859},
pages = {395-411} }

@article{roettgermeaningful,
title={Meaningful results for meaningful hypotheses: A tutorial on hypothesis testing with Bayes factors using ROPes},
author={Roettger, Timo and Franke, Michael},
publisher={OSF} }